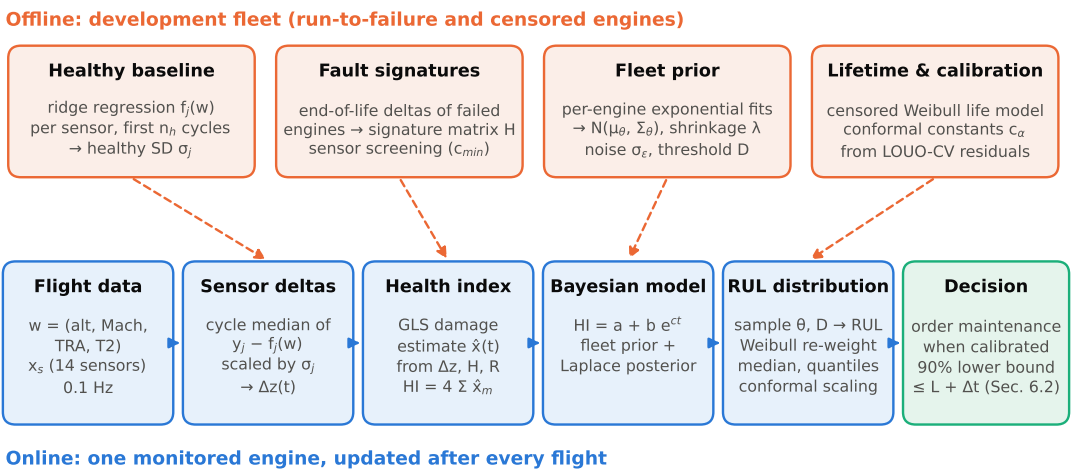
Offline: development fleet (run-to-failure and censored engines)
Healthy baseline
ridge regression $f_j(w)$
per sensor, first n_h cycles
→ healthy SD σ_j
Fault signatures
end-of-life deltas of failed
engines → signature matrix H
sensor screening (c_{min})
Fleet prior
per-engine exponential fits
→ $N(\mu_\theta, \Sigma_\theta)$, shrinkage λ
noise σ_ε, threshold D
Lifetime & calibration
censored Weibull life model
conformal constants c_α
from LOUO-CV residuals
Flight data
w = (alt, Mach,
TRA, T2)
x_s (14 sensors)
0.1 Hz
Sensor deltas
cycle median of
$y_j - f_j(w)$
scaled by σ_j
→ $\Delta z(t)$
Health index
GLS damage
estimate $\hat{x}(t)$
from Δz, H, R
HI = 4 Σ $\hat{x}_m$
Bayesian model
HI = a + b e^{ct}
fleet prior +
Laplace posterior
RUL distribution
sample θ, D → RUL
Weibull re-weight
median, quantiles
conformal scaling
Decision
order maintenance
when calibrated
90% lower bound
≤ L + Δt (Sec. 6.2)
Online: one monitored engine, updated after every flight